Coupled Transformer Framework for Thick Cloud Removal of Multitemporal Remote Sensing Images

Yang Zhou, Hao Peng, Chuan Wang, Teng-Yu Ji, *Member, IEEE*, Yong Chen, and Xi-Le Zhao, *Senior Member, IEEE*

***Abstract*—Multitemporal remote sensing images (MTRSIs) provide rich temporal information for Earth observation, but thick clouds severely hinder their practical applications. Supervised methods rely on paired cloudy/cloud-free training data, which limits their flexibility across different MTRSIs. Recently, unsupervised methods have attracted increasing attention because they do not require paired training data. However, their representation capability remains limited, particularly in modeling long-range spatiotemporal dependencies required in challenging scenarios with large cloud coverage. To address this challenge, we propose the first unsupervised coupled transformer framework (C-Transformer) for cloud removal, which offers flexible adaptability and long-range modeling ability. Going beyond a straightforward transplantation of Transformers to raw MTRSIs, the fundamental idea behind C-Transformer is to uncover and model the intrinsic spatiotemporal coherence hidden behind cloud-contaminated observations. Specifically, grounded in coupled tensor decomposition, C-Transformer disentangles an MTRSI into a coupled abundance tensor and uncoupled spectral signatures, and employs an untrained Transformer to implicitly represent the former, thereby capturing long-range dependencies while preserving temporal–spectral variability through spectral-guided deep decoders. Extensive experiments on both simulated and real-world datasets demonstrate that C-Transformer achieves the best average performance among competing methods both qualitatively and quantitatively (an average peak signal-to-noise ratio (PSNR) improvement of 2.6 dB).**

***Index Terms*—Coupled tensor decomposition, multitemporal remote sensing images (MTRSIs), thick cloud removal, unsupervised Transformer.**

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I. Introduction

With the rapid advancement of remote sensing technology, optical remote sensing images have become an indispensable data source for Earth observation, supporting critical applications such as environmental monitoring, urban planning, resource management, and agriculture [1], [2], [3], [4], [5]. However, the imaging process is inevitably affected by atmospheric disturbances, leading to frequent cloud contamination in the acquired images. In particular, thick clouds severely hinder the acquisition of continuous and complete cloud-free multitemporal observations by obscuring surface reflectance information and causing substantial spatial and spectral detail loss [6], [7], thereby undermining temporal consistency and the reliability of downstream analysis.

In recent years, various methods have been developed for thick cloud removal, aiming to recover cloud-contaminated regions while preserving the underlying image structure. Early studies mainly focused on single-temporal remote sensing images and exploited spatial [8], [9] and spectral information [10], [11] for reconstruction. However, severe cloud coverage often leaves insufficient information within a single observation [12], [13], [14], [15]. Multitemporal remote sensing images (MTRSIs) provide complementary spatial, spectral, and temporal information across different acquisition times, effectively alleviating this limitation. Accordingly, a series of methods [16], [17], [18], [19] have been developed to exploit the complementary information of MTRSIs for more accurate and robust cloud removal.

Existing MTRSI cloud removal methods can be broadly divided into supervised and unsupervised methods. Supervised methods learn mappings from cloud-contaminated observations to cloud-free images from training data and have achieved promising reconstruction performance [20], [21], [22]. However, because they are generally trained on paired cloudy/cloud-free data, their flexibility and generalization across MTRSIs acquired from different sensors and scenes are limited. More recently, unsupervised methods have attracted increasing attention due to their greater flexibility and generalization across different MTRSIs without requiring external training data.

For multitemporal observations, existing unsupervised methods generally follow two strategies: model-based methods with handcrafted priors and untrained deep methods with

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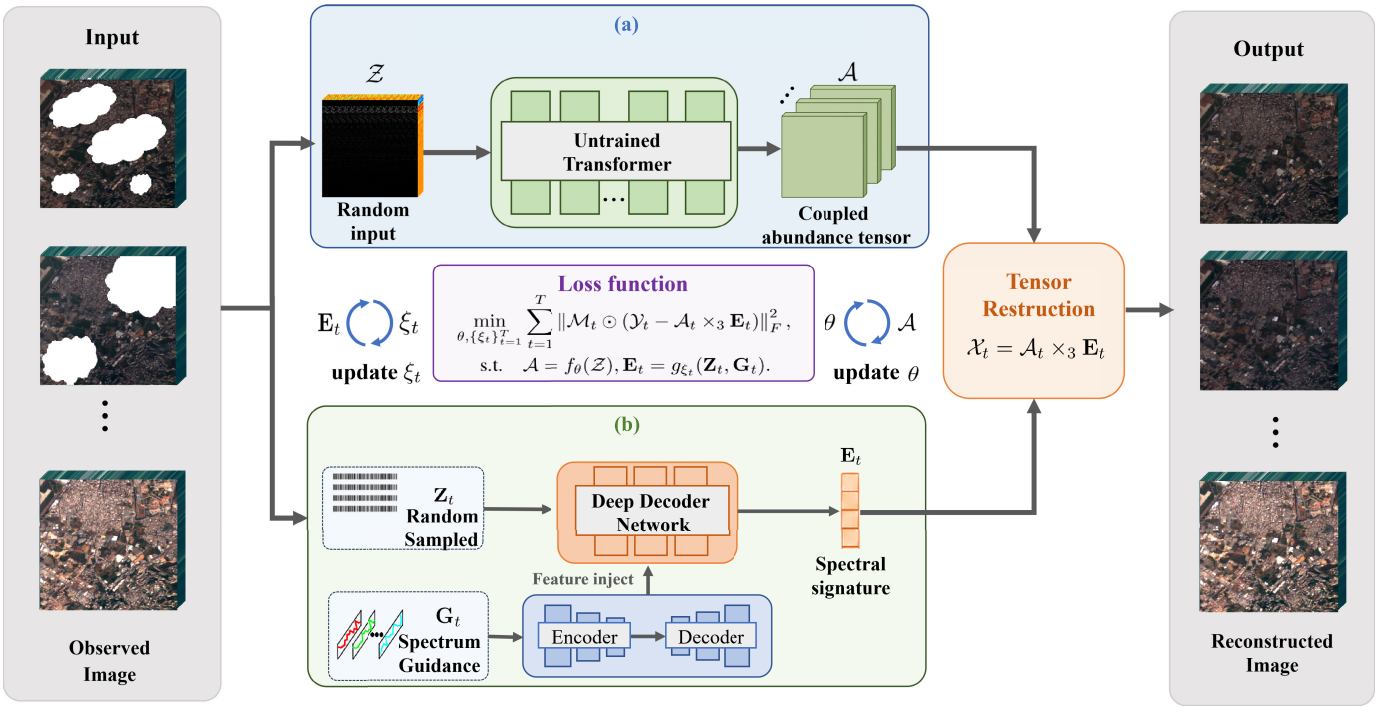


Fig. 1. Proposed C-Transformer represents MTRSIs through a coupled abundance tensor and time-specific spectral signatures. An untrained Transformer models the coupled abundance representation to capture long-range spatiotemporal dependencies, while spectral-guided decoders generate the corresponding spectral signatures. The two components are combined through tensor reconstruction, and the network parameters are updated using an observation fidelity constraint on reliable cloud-free pixels. (a) Transformer prior for abundance $f_\theta(\cdot)$. (b) Guided decoder prior for spectral $g_{\xi_t}(\cdot)$.

implicit neural priors. The first strategy incorporates handcrafted priors, such as low-rankness, sparsity, and smoothness [23], [24], [25], [26], into regularization terms to characterize the intrinsic structures of MTRSIs. However, such predefined priors often have limited representational capability for complex MTRSI structures. The second strategy exploits implicit neural priors, where the underlying structures are captured by untrained neural networks [27], [28], [29], [30], providing more flexible representations than handcrafted priors. Nevertheless, the network architectures adopted by these methods have limited capability to model long-range spatiotemporal dependencies, leading to degraded reconstruction performance, particularly under challenging scenarios with large cloud coverage. These limitations motivate the development of an untrained framework with long-range modeling capability and flexible adaptability for MTRSI thick cloud removal.

To alleviate these issues, we propose C-Transformer, an untrained coupled Transformer framework for MTRSI thick cloud removal. Compared with supervised methods, C-Transformer provides greater flexibility across different MTRSIs. Compared with existing unsupervised methods, it employs an untrained Transformer to capture long-range spatiotemporal dependencies, offering stronger modeling capability for complex MTRSI structures. Specifically, as shown in Fig. 1, each temporal image is decomposed into an abundance map and its corresponding spectral signature. The abundance maps of different temporal nodes are stacked to construct a coupled abundance tensor and represented by a single untrained Transformer. The uncoupled spectral signatures are represented by spectral-guided deep decoders to preserve temporal–spectral variability. Such coupling enables the joint exploitation of shared spatial structures and temporal correlations across different temporal nodes. Overall, C-Transformer combines the strong modeling capability of the Transformer with the flexibility of untrained learning, providing an effective framework for MTRSI thick cloud removal.

Our contributions are summarized as follows.

1) To address the limited flexibility of supervised methods and the weak representation capability of unsupervised methods, we propose the first unsupervised coupled Transformer framework (C-Transformer), which enjoys greater flexibility across different MTRSIs and stronger long-range modeling capability.
2) Beyond a straightforward transplantation of Transformers, C-Transformer uncovers the intrinsic spatiotemporal coherence hidden behind cloud-contaminated observations by decomposing an MTRSI into a coupled abundance tensor and uncoupled spectral signatures, with an untrained Transformer and spectral-guided deep decoders, respectively, representing the two components to capture long-range spatiotemporal dependencies and preserve temporal–spectral variability.
3) Extensive experiments on simulated and real-world MTRSI datasets demonstrate the effectiveness of C-Transformer. It achieves the best average performance among state-of-the-art methods with high structural consistency and spectral fidelity, particularly under large cloud coverage.

The remainder of this article is organized as follows. Section II reviews related work. Section III provides the main notations used in our work. Section IV details the proposed

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C-Transformer framework and its corresponding thick cloud removal model. Section V presents the experimental results on simulated and real-world datasets, followed by discussions. Finally, Section VI concludes this article.

II. Related Works

Early thick cloud removal methods mainly focused on single-temporal remote sensing images and reconstructed cloud-contaminated regions by exploiting spatial continuity and spectral correlation within the target image [13], [14]. With the development of deep learning, neural networks have further enhanced representation capability by learning spatial and spectral features from data [31], [32], while auxiliary observations have also been introduced to provide complementary information [33], [34]. However, single-temporal images may still provide insufficient information when large regions are completely obscured by thick clouds, especially in complex scenes with significant spatial variations. Therefore, MTRSIs have been increasingly exploited to provide complementary land-surface information from different acquisition times.

To further exploit such complementary information, increasingly advanced network architectures have been introduced for multitemporal cloud removal with the development of deep learning. Early deep learning methods mainly employed CNNs to extract and fuse complementary features from different temporal observations [20], [35], [36]. Subsequently, GAN-based methods were introduced to improve the generation of cloud-free images through adversarial learning [37]. More recently, Transformer-based methods have been developed to capture long-range dependencies among multitemporal images [38], [39], while diffusion models have further been explored to exploit multitemporal complementary information as conditional guidance for cloud-free image reconstruction [40], [41]. Despite their promising performance, their reliance on large-scale paired training data limits their flexibility across different MTRSIs. Due to these limitations, unsupervised multitemporal cloud removal methods have attracted increasing attention.

Early unsupervised methods mainly exploited temporal complementary information through statistical modeling, regression, sparse representation, and patch matching [16], [17], [24], [42]. For example, Tseng et al. [43] reconstructed cloud-contaminated regions using complementary information from multitemporal SPOT images, while Li et al. [44] exploited spectral–temporal correlations through sparse representation. Recently, CLEAR [45] established class-based linear regression relationships between temporal reference images and cloudy observations and further employed iterative residual compensation for cloud removal without model training.

Beyond these conventional methods, subsequent unsupervised methods can be broadly categorized into model-based methods with explicit priors and untrained deep methods with implicit neural priors. Model-based methods further exploit the intrinsic spatial–spectral–temporal structures of MTRSIs through handcrafted priors, such as low-rankness and sparsity [46], [47], [48], [49]. However, these predefined priors have limited capability to represent complex structures of MTRSIs. More recently, untrained deep methods have been introduced to provide more flexible nonlinear representations without paired training data [50], [51], [52]. For example, STLR-DP [53] combines low-rank tensor modeling with an untrained deep prior, while UNDFN [29] employs untrained neural networks to represent abundance maps and spectral signatures of multitemporal observations. Despite their improved representation capability, the performance of existing untrained deep methods remains limited by their insufficient long-range modeling capability, particularly under large cloud coverage. To address these limitations, C-Transformer represents abundance tensors across temporal nodes using an untrained Transformer and represents the corresponding spectral signatures with spectral-guided deep decoders. Compared with existing unsupervised methods, C-Transformer provides stronger long-range modeling capability for capturing complex spatiotemporal dependencies. Compared with supervised methods, it offers greater flexibility across different MTRSIs.

III. Preliminaries

In this section, we summarize the notations that are used in our work. Capital calligraphic letters (e.g., $\mathcal{X}$) and boldface capital letters (e.g., $\mathbf{X}$) denote tensors and matrices, respectively. For an mth order tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times \ldots \times n_m}$, its $(i_1, i_2, \ldots, i_m)$th element is represented by $\mathcal{X}_{i_1,i_2,\ldots,i_m}$. The Frobenius norm of $\mathcal{X}$ is defined as $\|\mathcal{X}\|_F = \left(\sum_{i_1,i_2,\ldots,i_m} |\mathcal{X}_{i_1,i_2,\ldots,i_m}|^2\right)^{1/2}$. For $\{\mathcal{A}_j\}_{j=1}^{n_4}$ where $\mathcal{A}_j \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we use $\mathcal{A} = \mathscr{L}(\{\mathcal{A}_j\}_{j=1}^{n_4}) \in \mathbb{R}^{n_1 \times n_2 \times n_3 \times n_4}$ to denote the stacking of all $\mathcal{A}_j$ along the fourth mode.

Definition 1 (k-Mode Tensor-Matrix Product [54]): The k-mode tensor–matrix product of a tensor $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times \ldots \times n_m}$ and a matrix $\mathbf{U} \in \mathbb{R}^{J \times n_k}$ is denoted by $\mathcal{A} \times_k \mathbf{U}$ and satisfies

$$(\mathcal{A} \times_k \mathbf{U})_{i_1,\ldots,i_{k-1},j,i_{k+1},\ldots,i_m} = \sum_{i_k=1}^{n_k} a_{i_1,i_2,\ldots,i_m} u_{j,i_k}.$$

IV. Proposed Method

In this section, we introduce the proposed unsupervised coupled Transformer framework (C-Transformer) for MTRSIs. Under C-Transformer, we build the corresponding thick cloud removal model.

A. Overview of the Proposed C-Transformer

For a multitemporal cloud-free remote sensing image $\mathcal{X} \in \mathbb{R}^{H \times W \times C \times T}$, where H and W denote the spatial dimensions, C denotes the spectral dimension, and T denotes the number of temporal nodes, the image at the tth temporal node $\mathcal{X}_t \in \mathbb{R}^{H \times W \times C}$ can be decomposed based on the linear spectral unmixing model [55] as

$$\mathcal{X}_t = \mathcal{A}_t \times_3 \mathbf{E}_t \tag{1}$$

where $\mathcal{A}_t \in \mathbb{R}^{H \times W \times r}$ denotes the abundance maps, $\mathbf{E}_t \in \mathbb{R}^{C \times r}$ denotes the corresponding spectral signature, and r is the number of abundance components. Here, $\times_3$ denotes the mode-3 tensor-matrix product.

Based on the above decomposition, we observe that the abundance maps across temporal observations exhibit intrinsic structural consistency, as the underlying land-cover distributions generally share substantial spatial structures over

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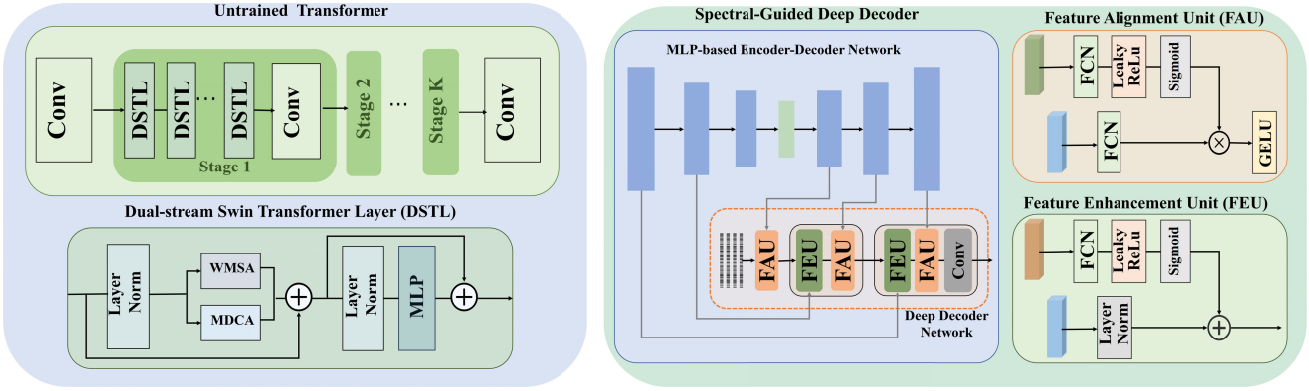


Fig. 2. Detailed network structures of the proposed C-Transformer. (a) Untrained coupled Transformer consists of convolutional projection layers and stacked DSTLs, where WMSA models spatial dependencies and MDCA enhances channel-wise interactions in the coupled abundance domain. (b) Spectral-guided deep decoder consists of an MLP-based encoder–decoder network and a deep decoder network with feature alignment units (FAUs) and feature enhancement units (FEUs), which use coarse spectral guidance to generate uncoupled spectral signatures.

short temporal intervals, whereas the spectral signatures may vary over time. Motivated by these observations, we propose C-Transformer, an untrained coupled Transformer framework for MTRSIs. Specifically, the abundance maps of all temporal nodes are stacked to form a coupled abundance tensor $\mathcal{A} = \mathscr{L}(\{\mathcal{A}_t\}_{t=1}^{T}) \in \mathbb{R}^{H\times W\times r\times T}$. To capture long-range spatial dependencies and temporal interactions among abundance maps, a single untrained Transformer is employed to represent the coupled abundance tensor. To preserve time-specific spectral variations, spectral-guided deep decoders are employed to independently represent the corresponding spectral signatures. The detailed representations of the coupled abundance tensor and spectral signatures are described as follows.

1) Untrained Transformer for Abundance Modeling: To represent the coupled abundance tensor $\mathcal{A}$, we employ an untrained Transformer f_θ, which maps a fixed random input $\mathcal{Z} \in \mathbb{R}^{H\times W\times rT}$ to

$$\mathcal{A} = f_\theta(\mathcal{Z}) \in \mathbb{R}^{H\times W\times r\times T} \tag{2}$$

where θ denotes the learnable parameters, and the untrained Transformer imposes an implicit prior on $\mathcal{A}$. The coupled abundance tensor $\mathcal{A}$ is constructed by stacking the abundance maps of all temporal nodes, where $\mathcal{A}_t$ denotes the abundance representation associated with the tth temporal node. This joint modeling captures long-range spatial dependencies within individual observations and exploits structural consistency across temporal nodes.

In this work, f_θ is instantiated with a staged Transformer architecture, as illustrated in Fig. 2(a). Convolutional layers are first employed for feature projection, followed by stacked dual-stream Swin Transformer [56] layers (DSTLs) [56] for feature representation. Each DSTL consists of a shifted-window multihead self-attention (WMSA) branch for spatial dependency modeling and a multihead dynamic channel attention (MDCA) branch [57] for interactions among coupled abundance components from different temporal nodes. This dual-stream architecture therefore enables joint modeling of spatial dependencies and cross-temporal interactions in the coupled abundance tensor.

2) Untrained Spectral-Guided Decoder for Spectral Signature Modeling: To represent the spectral component, we employ an independent untrained spectral-guided decoder g_{ξ_t} for each temporal node. For the tth temporal node, the decoder takes a fixed random input $\mathbf{Z}_t$ together with coarse spectral guidance $\mathbf{G}_t$ and represents the corresponding spectral signature as

$$\mathbf{E}_t = g_{\xi_t}(\mathbf{Z}_t, \mathbf{G}_t) \in \mathbb{R}^{C\times r} \tag{3}$$

where ξ_t denotes the learnable parameters of the decoder. Here, $\mathbf{Z}_t$ serves as the fixed random input, while $\mathbf{G}_t$ is extracted from reliable cloud-free pixels through mask pooling and a lightweight MLP.

As shown in Fig. 2(b), each g_{ξ_t} is instantiated using the parallel guided architecture of guided deep decoder [58]. To adapt this architecture from spatial image restoration to spectral signature modeling, the convolutional operators are replaced with fully connected layers, allowing the decoder to directly model dependencies among spectral bands.

B. C-Transformer for MTRSI Thick Cloud Removal

Based on the proposed C-Transformer framework, we further formulate the thick cloud removal model for MTRSIs. For a cloud-contaminated MTRSI $\mathcal{Y} \in \mathbb{R}^{H\times W\times C\times T}$, the degradation model at the tth temporal node is written as

$$\mathcal{Y}_t = \mathcal{M}_t \odot \mathcal{X}_t, \quad t = 1, \ldots, T \tag{4}$$

where $\odot$ denotes the Hadamard product, $\mathcal{X}_t \in \mathbb{R}^{H\times W\times C}$ denotes the cloud-free image at the tth temporal node, and $\mathcal{M}_t \in \mathbb{R}^{H\times W\times C}$ is the corresponding binary cloud mask. Entries with value 0 correspond to cloud-contaminated regions, while entries with value 1 correspond to cloud-free observations.

Under the C-Transformer framework, a single untrained Transformer f_θ is employed to represent the coupled abundance tensor $\mathcal{A} \in \mathbb{R}^{H\times W\times r\times T}$, while the spectral signature $\mathbf{E}_t \in \mathbb{R}^{C\times r}$ at each temporal node is represented by the

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Algorithm 1 Optimization Algorithm for C-Transformer

Input: Observed MTRSI $\mathcal{Y}$, cloud mask $\mathcal{M}$, maximum iteration number $k_{\max}$, and rank r. $\{\mathbf{Z}_t\}_{t=1}^T$ and $\mathcal{Z}$ are sampled from $\mathcal{U}(0,1)$ with seed 0, and the coarse spectral guidance $\{\mathbf{G}_t\}_{t=1}^T$, obtained by mask pooling cloud-free pixels and mapping the pooled spectrum to a $C \times r$ matrix using a lightweight MLP.

Initialize $k = 0$, $k_{\max} = 15000$, $\theta^{(0)}$ and $\{\xi_t^{(0)}\}_{t=1}^T$ using the default normal distribution in PyTorch.
while not converged and $k < k_{\max}$ **do**
  $\mathcal{A} = f_\theta^{(k)}(\mathcal{Z}), \mathcal{A}_t = \mathcal{A}(:,:,:,t)$ for $t = 1, 2, \ldots, T$.
  for $t = 1, 2, \ldots, T$ **do**
  $\mathbf{E}_t = g_{\xi_t}^{(k)}(\mathbf{Z}_t, \mathbf{G}_t)$.
  end for
  Compute gradients w.r.t. θ and $\{\xi_t\}_{t=1}^T$.
  Update $\theta^{(k+1)}$ and $\{\xi_t^{(k+1)}\}_{t=1}^T$ using Adam.
  $k \leftarrow k + 1$.
end while
$\mathcal{X}_t = (1 - \mathcal{M}_t) \odot (\mathcal{A}_t \times_3 \mathbf{E}_t) + \mathcal{M}_t \odot \mathcal{Y}_t$ for $t = 1, 2, \ldots, T$.

Output: The reconstructed image $\mathcal{X}$.

corresponding spectral-guided decoder g_{ξ_t}. Accordingly, the thick cloud removal model is formulated as

$$\begin{aligned} \min_{\theta, \{\xi_t\}_{t=1}^T} \quad & \sum_{t=1}^{T} \|\mathcal{M}_t \odot (\mathcal{Y}_t - \mathcal{X}_t)\|_F^2 \\ \text{s.t.} \quad & \mathcal{A} = f_\theta(\mathcal{Z}), \ \mathbf{E}_t = g_{\xi_t}(\mathbf{Z}_t, \mathbf{G}_t) \\ & \mathcal{A}_t = \mathcal{A}(:,:,:,t), \ \mathcal{X}_t = \mathcal{A}_t \times_3 \mathbf{E}_t. \end{aligned} \tag{5}$$

Here, $\mathcal{Z} \in \mathbb{R}^{H \times W \times rT}$ and $\mathbf{Z}_t \in \mathbb{R}^{C \times r}$ are fixed random inputs, while $\mathbf{G}_t \in \mathbb{R}^{C \times r}$ denotes the coarse spectral guidance for the tth temporal node.

Accordingly, the resulting optimization problem can be written as follows:

$$\begin{aligned} \min_{\theta, \{\xi_t\}_{t=1}^T} \quad & \sum_{t=1}^{T} \|\mathcal{M}_t \odot (\mathcal{Y}_t - \mathcal{A}_t \times_3 \mathbf{E}_t)\|_F^2 \\ \text{s.t.} \quad & \mathcal{A} = f_\theta(\mathcal{Z}) \\ & \mathbf{E}_t = g_{\xi_t}(\mathbf{Z}_t, \mathbf{G}_t). \end{aligned} \tag{6}$$

The overall workflow of the proposed C-Transformer for MTRSI thick cloud removal is illustrated in Fig. 1.

C. Solving Algorithm

The C-Transformer framework optimization is summarized in Algorithm 1. Specifically, we employ the adaptive moment estimation (Adam) [59] optimizer to tackle problem (6), which is particularly tailored for optimizing deep neural networks. Meanwhile, we set a maximum iteration number $k_{\max}$ as the stopping criterion of the Adam optimization. The estimates of the coupled abundance tensor and the spectral signatures are updated by $\mathcal{A} = f_\theta(\mathcal{Z})$ and $\mathbf{E}_t = g_{\xi_t}(\mathbf{Z}_t, \mathbf{G}_t)$, and the reconstructed image at the tth temporal node is obtained by calculating $\mathcal{X}_t = (1 - \mathcal{M}_t) \odot (\mathcal{A}_t \times_3 \mathbf{E}_t) + \mathcal{M}_t \odot \mathcal{Y}_t$, where $t = 1, 2, \ldots, T$.

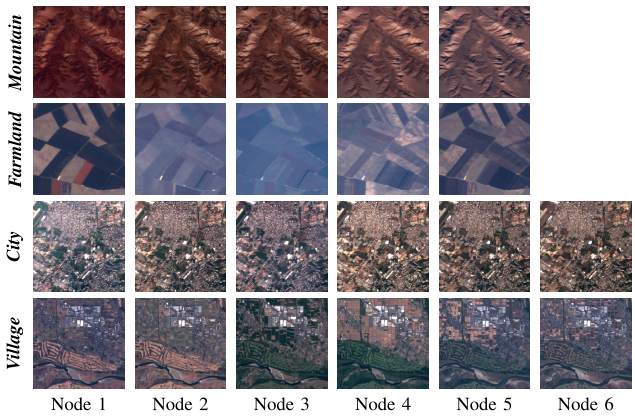


Fig. 3. Visual display (R:3, G:2, B:1) of MTRSIs for the simulated datasets *mountain*, *farmland*, *city*, and *village*.

TABLE I
DESCRIPTION OF THE FOUR SIMULATED DATASETS

Dataset	Source	Image size	Spectral bands	Temporal nodes	Spatial resolution
Mountain	Sentinel-2	256×256	4	5	10 m
Farmland	Sentinel-2	256×256	8	5	10 m
City	Sentinel-2	256×256	4	6	10 m
Village	Sentinel-2	256×256	4	6	10 m

V. EXPERIMENTS AND RESULTS

In this section, we evaluate C-Transformer on simulated and real-world data and compare it with six state-of-the-art methods, including HaLRTC [60], TVTR [48], RTCR [49], UNDFN [29], STLR-DP [53], and CLEAR [45]. The comparison code is obtained from the authors or official repositories, with parameters tuned following the corresponding literature. For numerical stability and visualization, each spectral band of the MTRSIs is normalized to $[0, 1]$ before the experiments. All experiments are implemented using a Windows 11 platform with one Intel Core i9-9900K CPU at 3.60 GHz, 32 GB RAM, and an NVIDIA RTX PRO 6000 GPU. The implementations of CLEAR, STLR-DP, UNDFN, and C-Transformer are conducted in PyTorch 2.8.0, while the other comparison methods are conducted in MATLAB R2021b.

A. Simulated Experiments

As the quantitative assessment of thick cloud removal in real scenarios remains challenging, we employed simulated experiments to verify the efficacy of the proposed C-Transformer. Detailed information regarding the simulated datasets and evaluation metrics is provided below.

1) Datasets: To verify the flexibility of the proposed method under different land-cover scenarios, we selected three typical scenes from the SEN12MS-CR-TS[1] dataset [27], termed *mountain*, *farmland*, and *city*. In addition, a *village* scene was extracted from the Sentinel-2[2] dataset, resulting in four distinct test samples. Notably, the *village* scene exhibits noticeable land-cover variations across

[1]https://github.com/PatrickTUM/SEN12MS-CR-TS
[2]https://dataspace.copernicus.eu

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TABLE II

QUANTITATIVE COMPARISONS ON DIFFERENT DATASETS AND CONTAMINATED NODES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Region	Time nodes	Metric	Observed	Regression	HaLRTC	TVTR	RTCR	UNDFN	STLR–DP	CLEAR	C-Transformer
Mountain	Node 1	PSNR	7.322	32.635	33.966	36.842	40.234	42.829	43.065	35.818	**45.225**
		SSIM	0.6992	0.9100	0.9580	0.9403	0.9774	0.9823	0.9840	0.9327	**0.9903**
		SAM	11.7708	1.0888	2.1080	1.0913	0.5404	0.5261	0.5040	0.8864	**0.4602**
	Node 2	PSNR	7.501	33.465	32.910	36.332	37.804	43.182	41.355	37.478	**47.406**
		SSIM	0.6810	0.9481	0.9470	0.9399	0.9714	0.9829	0.9799	0.9609	**0.9935**
		SAM	10.6127	1.4018	1.5906	1.1343	0.6695	0.5230	0.5019	0.8669	**0.3804**
	Node 3	PSNR	9.578	36.592	38.655	38.531	42.794	46.030	46.497	38.626	**49.734**
		SSIM	0.8014	0.9526	0.9726	0.9546	0.9837	0.9894	0.9904	0.9596	**0.9954**
		SAM	6.7958	0.7558	0.7601	0.6905	0.3403	0.3134	0.2848	0.6039	**0.1985**
Farmland	Node 1	PSNR	8.543	28.304	25.803	29.322	29.722	35.061	35.963	34.394	**38.026**
		SSIM	0.7524	0.9027	0.9128	0.8889	0.9269	0.9506	0.9669	0.9476	**0.9745**
		SAM	10.8740	2.7282	3.2667	2.3239	2.4272	0.9766	0.8639	0.9492	**0.6095**
	Node 2	PSNR	9.143	26.803	28.939	30.448	29.679	32.795	**37.225**	34.204	36.987
		SSIM	0.7195	0.8817	0.9279	0.8842	0.9211	0.9268	0.9510	0.9206	**0.9570**
		SAM	8.2389	3.1295	1.5759	2.4775	2.0993	1.5363	**0.7498**	1.0396	0.7890
	Node 3	PSNR	10.034	31.221	31.397	30.525	32.670	35.608	40.752	38.626	**42.101**
		SSIM	0.8295	0.9510	0.9588	0.9095	0.9579	0.9581	0.9747	0.9591	**0.9783**
		SAM	4.1874	1.5903	1.3813	3.0698	1.6282	1.2562	0.5980	0.6124	**0.4797**
City	Node 1	PSNR	7.945	28.795	25.302	30.600	31.534	33.069	30.842	30.170	**33.917**
		SSIM	0.6802	0.8907	0.9078	0.9107	0.9529	0.9597	0.9482	0.9099	**0.9660**
		SAM	4.8509	1.7130	1.8612	1.7039	1.1774	1.1226	1.1942	1.6194	**1.0769**
	Node 2	PSNR	7.820	30.093	29.835	32.363	34.687	35.646	34.341	32.629	**36.404**
		SSIM	0.6479	0.8958	0.9348	0.9235	0.9655	0.9706	0.9682	0.9349	**0.9745**
		SAM	5.6827	1.5723	1.4386	1.4153	0.9004	0.8881	0.8990	1.3191	**0.8230**
	Node 3	PSNR	9.514	30.235	29.132	31.490	32.409	36.103	34.660	32.651	**37.406**
		SSIM	0.7864	0.8974	0.9331	0.9202	0.9517	0.9727	0.9632	0.9353	**0.9784**
		SAM	4.0997	1.3610	1.1773	1.1546	0.8428	0.6942	0.7721	1.1254	**0.5952**
Village	Node 1	PSNR	6.853	30.776	28.149	32.901	33.995	34.979	33.668	32.986	**36.888**
		SSIM	0.6898	0.8947	0.8562	0.9133	0.9428	0.9457	0.9429	0.9198	**0.9619**
		SAM	4.5347	2.2508	2.7159	2.2220	1.4879	1.4092	1.2902	1.6762	**1.0623**
	Node 2	PSNR	6.587	28.793	27.721	32.838	34.470	35.740	35.829	32.335	**37.785**
		SSIM	0.5849	0.8546	0.8200	0.9000	0.9461	0.9441	0.9534	0.8980	**0.9628**
		SAM	6.1851	3.3024	2.7455	2.1636	1.5644	1.3178	1.1616	1.8911	**1.0227**
	Node 3	PSNR	9.395	30.564	30.666	32.143	33.268	35.849	35.180	33.136	**38.455**
		SSIM	0.7856	0.9063	0.8972	0.9114	0.9414	0.9521	0.9569	0.9264	**0.9690**
		SAM	7.2544	2.0641	2.1708	1.7988	1.6069	1.0148	0.9825	1.4085	**0.7327**
	Node 4	PSNR	7.327	27.249	25.177	28.753	29.477	33.130	31.902	28.751	**35.347**
		SSIM	0.6722	0.8336	0.8372	0.8360	0.9010	0.9172	0.9207	0.8566	**0.9448**
		SAM	7.3755	2.9545	3.7944	3.4277	2.7861	1.5989	1.5126	2.1446	**1.1885**
Average		PSNR	8.274	30.425	29.819	32.545	34.057	36.925	37.021	33.985	**39.668**
		SSIM	0.7177	0.9015	0.9126	0.9102	0.9492	0.9579	0.9616	0.9278	**0.9728**
		SAM	7.1125	1.9933	2.0451	1.8979	1.3901	1.0136	0.8704	1.2417	**0.7245**

different temporal nodes, providing a more challenging setting for thick cloud removal in MTRSIs. As summarized in Table I and Fig. 3, these datasets cover different land-cover types, spectral bands, and temporal nodes, providing a comprehensive evaluation setting for MTRSI thick cloud removal.

In the simulated experiments, we introduced simulated thick clouds with a cloud coverage ratio of 40%–60%, forming challenging large-cloud removal scenarios. Specifically, the *mountain* and *farmland* data each contain three cloudy and two cloud-free temporal nodes, the *village* data contain four cloudy and two cloud-free temporal nodes, and the *city* data contain three cloudy and three cloud-free temporal nodes. Simulated clouds vary in location, shape, and size across temporal nodes, while sharing the same spatial mask across spectral bands within each node. This setting enables evaluation under severe cloud contamination across different landscapes.

2) Evaluation Indexes: To quantitatively evaluate the performance of different thick cloud removal methods, we employed three commonly used quantitative metrics: peak signal-to-noise ratio (PSNR) [61], structural similarity index (SSIM) [61], and spectral angle mapper (SAM) [62]. Specifically, higher PSNR and SSIM values indicate superior reconstruction performance, reflecting better radiometric accuracy and structural fidelity, respectively. Conversely, the SAM measures the spectral angle between the reconstructed image

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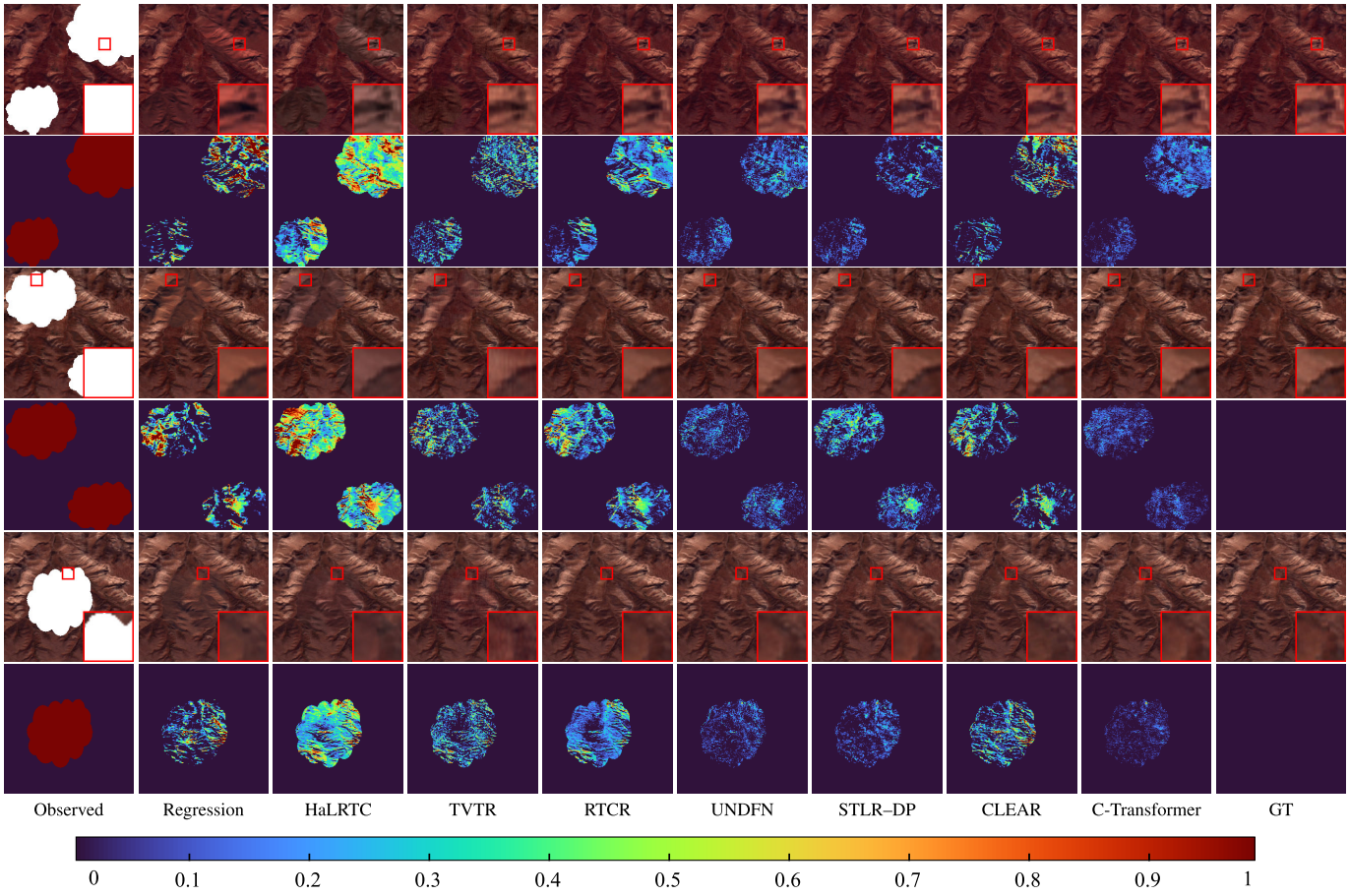


Fig. 4. Experimental results of various methods on the *mountain* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, and 3. Odd rows show the RGB composites (R:3, G:2, B:1), and even rows show the residual maps obtained by averaging residuals over all bands.

TABLE III
DESCRIPTION OF THE TWO REAL DATASETS

Dataset	Source	Image size	Spectral bands	Temporal nodes	Spatial resolution
Mantoval	Sentinel-2	256×256	4	4	10 m
Bourgogne	Landsat 8	256×256	7	4	30 m

and the ground truth (GT). A lower SAM value signifies a higher spectral similarity to the original image.

3) Quantitative Comparison: Table II reports the quantitative results on four datasets. C-Transformer achieves the best average performance, improving the average PSNR by 2.6 dB over the second-best method while achieving the highest SSIM on all datasets. Model-based methods such as HaLRTC, TVTR, and RTCR achieve reasonable restoration by exploiting handcrafted priors, but their limited representation capability restricts the modeling of complex spatial and temporal structures. CLEAR preserves local structures and edge details through its clustering-regression strategy, but its performance depends on the available temporal references. UNDFN and STLR-DP improve reconstruction by combining low-rank and deep priors, yet remain limited in long-range dependency modeling. The superior performance of C-Transformer benefits from coupled abundance modeling and the strong representation capability of the Transformer. Notably, C-Transformer still achieves favorable performance on the *village* dataset with substantial land-cover variations, demonstrating that it remains suitable for MTRSIs with changing land-cover conditions.

4) Visual Comparison: For visual comparison, Figs. 4–7 present the reconstruction results of different methods with the GT. The regions marked by red rectangles are enlarged in the bottom-right corner, and residual maps are shown below the reconstructed images. As observed, RTCR preserves the overall multitemporal structures through coupled tensor decomposition, but some fine details remain blurred. CLEAR achieves better local structure and edge recovery through its clustering-regression strategy. UNDFN and STLR-DP further improve visual quality by introducing deep priors, but noticeable artifacts and spectral deviations remain in cloud-covered regions. In contrast, C-Transformer reconstructs clearer spatial details with smaller residual errors and better spectral consistency. This improvement benefits from the effective exploitation of structural information across multiple temporal observations and the strong representation capability of the Transformer, while the spectral-guided decoders further exploit the available spectral information to better represent fine details.

To further evaluate radiometric consistency in cloud-contaminated regions, Fig. 8 presents scatter plots between the

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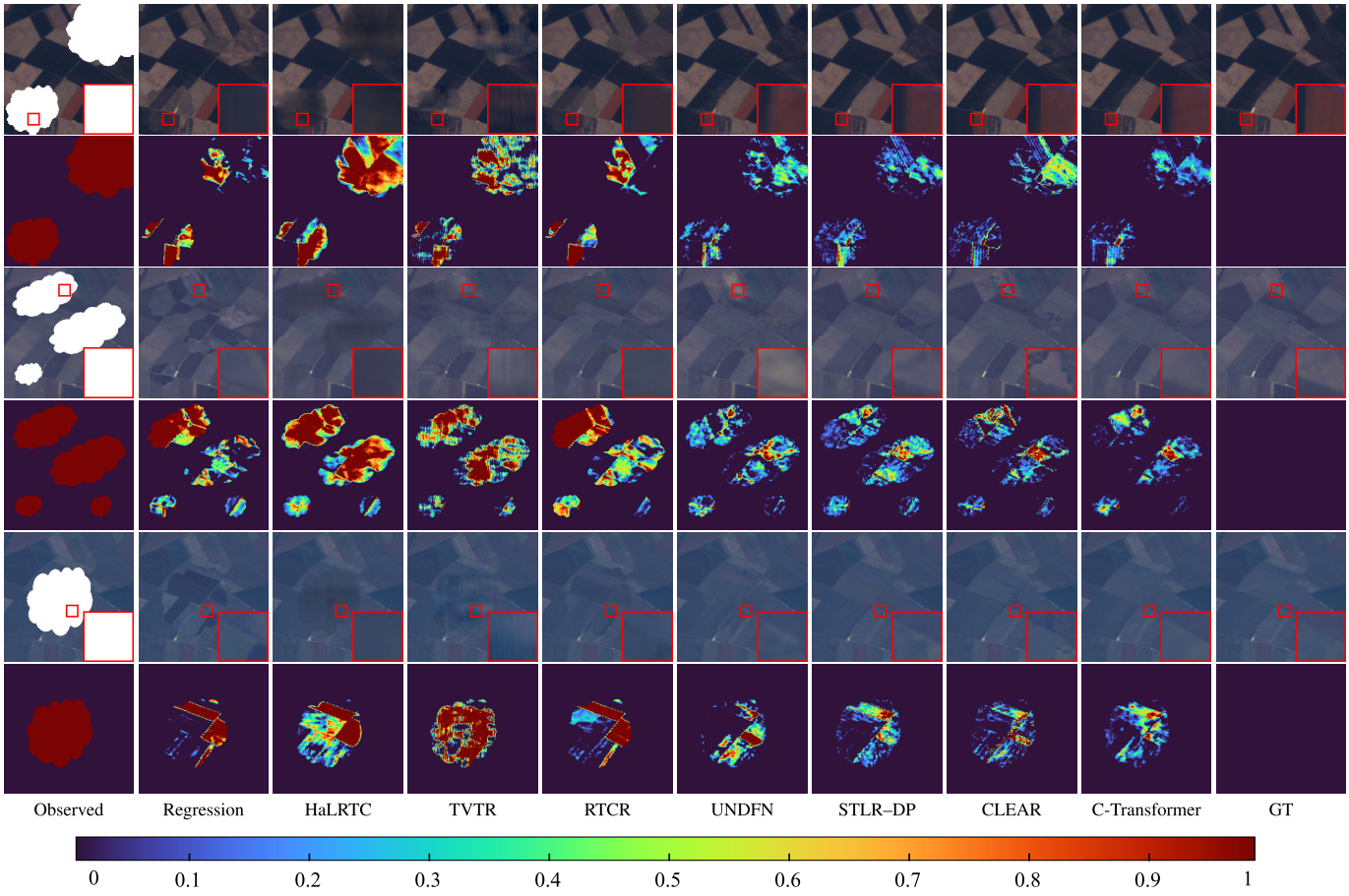


Fig. 5. Experimental results of various methods on the *farmland* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, and 3. Odd rows show the RGB composites (R:3, G:2, B:1), and even rows show the residual maps obtained by averaging residuals over all bands.

GT and reconstructed pixel values under different scenarios. The horizontal and vertical axes denote the GT and reconstructed pixel values, respectively, where a tighter distribution around the diagonal indicates higher reconstruction accuracy. As shown in Fig. 8, most methods perform relatively well on the *city* and *mountain* datasets. However, on the more challenging *farmland* and *village* datasets, HaLRTC, TVTR, CLEAR, and RTCR exhibit obvious deviations from the diagonal, indicating limited radiometric fidelity. Although STLR-DP produces more concentrated scatter points than other methods, noticeable deviations remain. In contrast, the scatter points of C-Transformer are more tightly distributed around the diagonal, demonstrating better pixel-level fidelity.

B. Real Experiments

To provide a more comprehensive evaluation, we conduct experiments on two real-world datasets from different satellites, namely, *Mantoval* and *Bourgogne*. Since the original images are large, cloud-contaminated subimages with a spatial size of 256×256 are cropped for testing. Detailed information is summarized in Table III, and representative examples are shown in Fig. 9. The *Mantoval* dataset was acquired by Sentinel-2, where cloud contamination occurs in the cropped region at the first and third temporal nodes; the data and corresponding cloud masks are available from the Copernicus Data Space Ecosystem website.[3] The *Bourgogne* dataset was acquired by Landsat 8, with cloud contamination in the cropped region at the first and second temporal nodes; both the data and masks were obtained from the website.[4] The remaining cloud-free observations are used as reference images to support multitemporal reconstruction.

For visual comparison, Figs. 10 and 11 present the reconstruction results of different thick-cloud removal methods. The red rectangular regions are magnified to highlight local differences. As observed, HaLRTC and TVTR show limited ability to recover fine structures, while RTCR preserves the overall scene layout but produces less refined details. CLEAR effectively preserves local structures and edge details through its clustering-regression strategy. UNDFN and STLR-DP improve edge and texture preservation, but still suffer from spectral inconsistency in some regions. In contrast, C-Transformer produces more consistent reconstructions with clearer spatial structures and better spectral fidelity, benefiting from the long-range modeling capability of the untrained coupled Transformer and the effective exploitation of available spectral information by the spectral-guided deep decoders.

C. Discussion

[3]https://dataspace.copernicus.eu
[4]https://theia.cnes.fr/atdistrib/rocket/#/home

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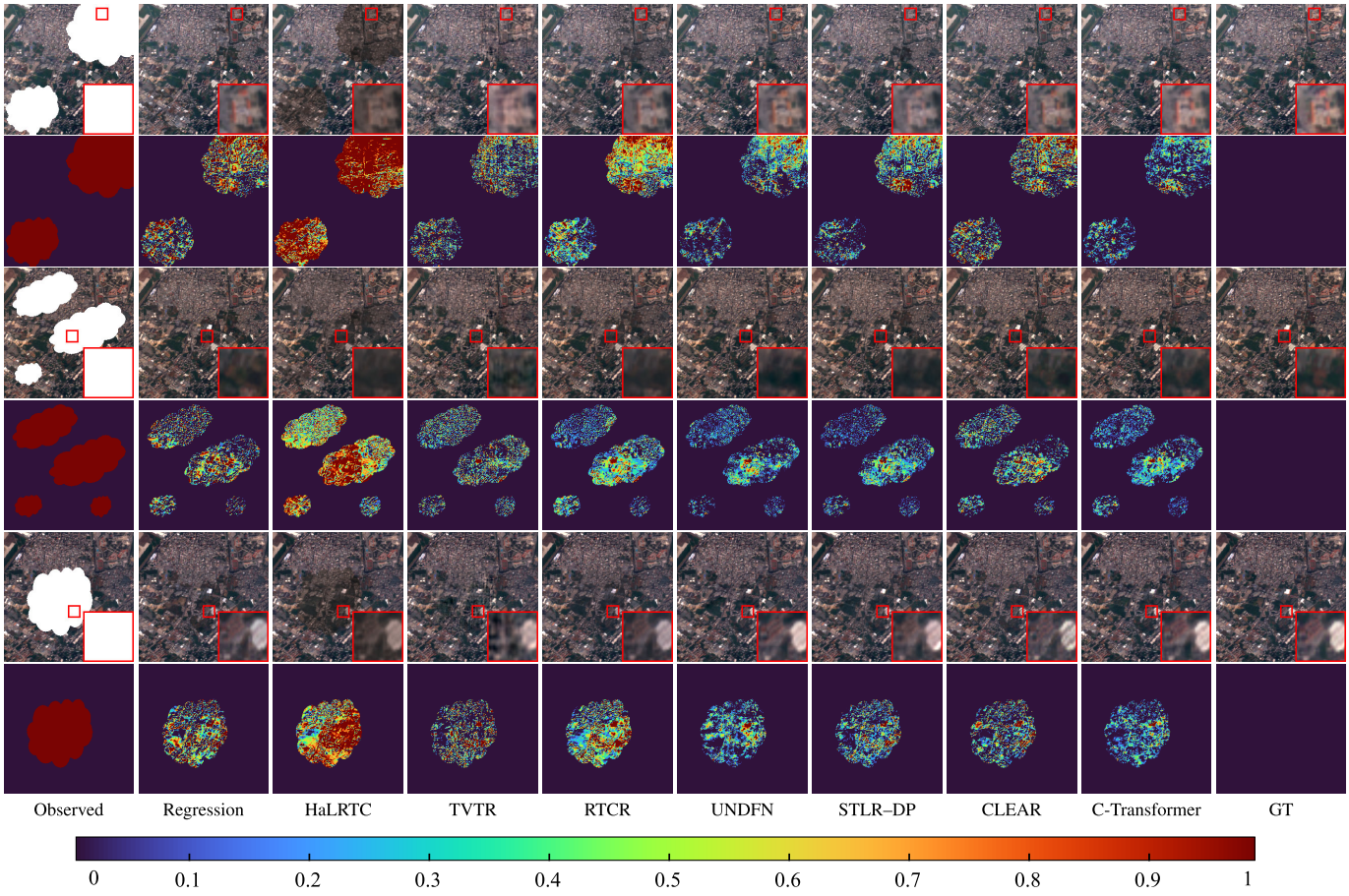


Fig. 6. Experimental results of various methods on the *city* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, and 3. Odd rows show the RGB composites (R:3, G:2, B:1), and even rows show the residual maps obtained by averaging residuals over all bands.

TABLE IV

QUANTITATIVE COMPARISON UNDER DIFFERENT CLOUD COVERAGE RATIOS. THE BEST RESULTS ARE IN BOLD

Cloud coverage	Metric	CNN	UNet	C-Transformer
20%	MPSNR	44.523	46.607	**48.373**
	MSSIM	0.9885	0.9928	**0.9949**
	MSAM	0.3414	0.2906	**0.2301**
40%	MPSNR	40.499	42.490	**43.821**
	MSSIM	0.9736	0.9825	**0.9865**
	MSAM	0.7450	0.6088	**0.5177**
60%	MPSNR	36.521	38.963	**40.215**
	MSSIM	0.9390	0.9628	**0.9712**
	MSAM	1.4012	1.0576	**0.9051**
80%	MPSNR	35.442	35.914	**36.979**
	MSSIM	0.9227	0.9320	**0.9441**
	MSAM	1.7478	1.6984	**1.4514**

TABLE V

CONTRIBUTION OF THE ABUNDANCE MODELING MECHANISM. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Metric	Uncoupled Transformer	Coupled Transformer
MPSNR	30.202	**47.455**
MSSIM	0.8797	**0.9930**
MSAM	1.5026	**0.3463**
Params	5.05M	**1.18M**
GPU Memory	42.6GB	**9.1GB**

1) Power of Long-Range Dependency Modeling: To evaluate the effect of long-range dependency modeling, we compare CNN, UNet, and Swin Transformer backbones with comparable parameter scales for coupled abundance modeling under different cloud coverage ratios. The results on the *mountain* dataset are reported in Table IV using mean PSNR (MPSNR), mean SSIM (MSSIM), and mean SAM (MSAM), where higher MPSNR and MSSIM and lower MSAM indicate better reconstruction performance. As cloud coverage increases, all methods exhibit performance degradation, while CNN-based architectures deteriorate more severely due to their limited long-range modeling capability. In contrast, C-Transformer achieves superior performance with a slower degradation trend, demonstrating the effectiveness of long-range dependency modeling for large cloud-covered regions.

2) Contribution of the Coupled Mechanism: To validate the coupled mechanism, we compare two abundance modeling strategies: uncoupled and coupled. The uncoupled strategy reconstructs each temporal abundance tensor with a separate untrained Transformer, leading to independent optimization

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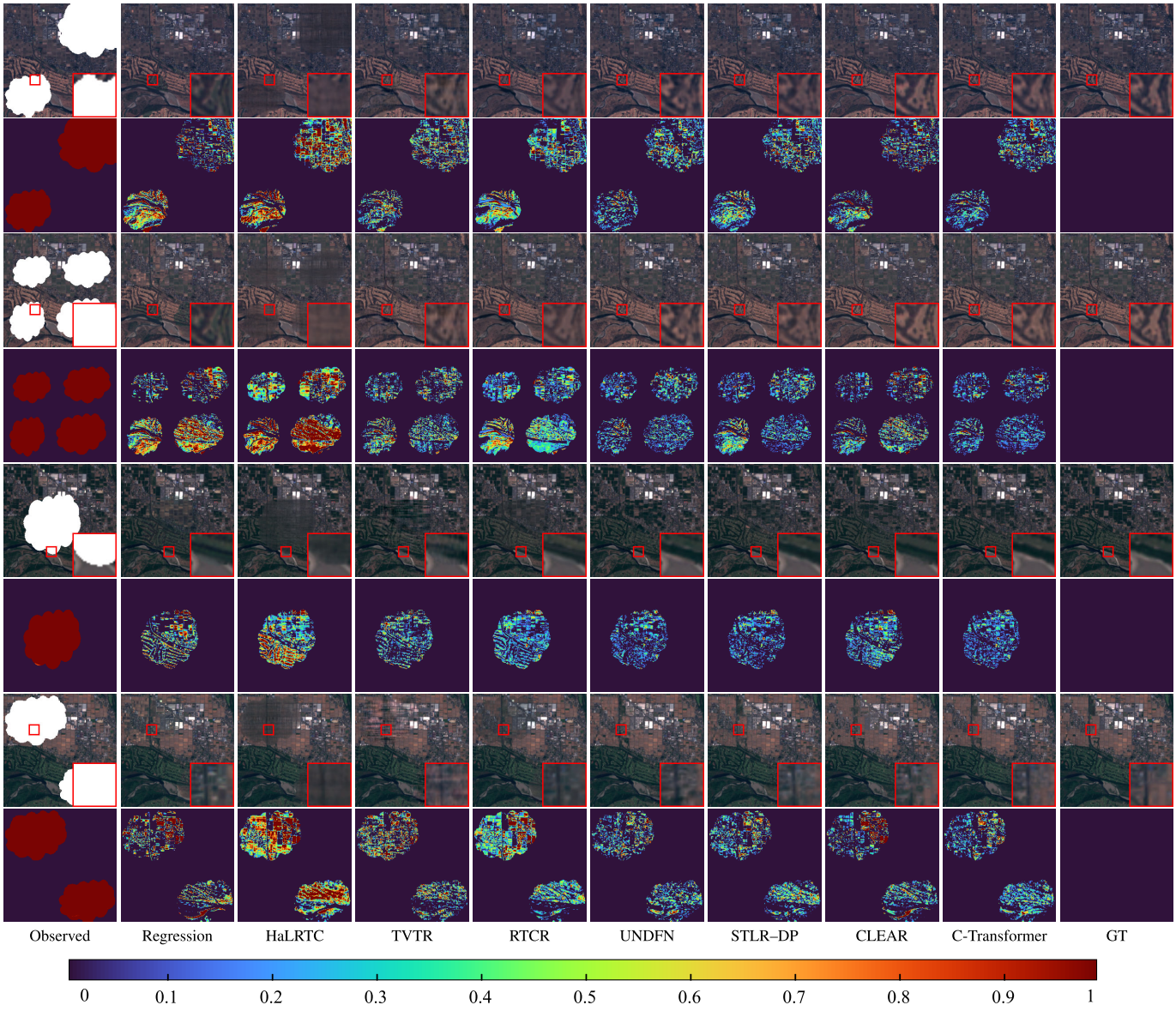


Fig. 7. Experimental results of various methods on the *village* dataset. The images are presented from top to bottom at contaminated nodes 1, 2, 3, and 4. Odd rows show the RGB composites (R:3, G:2, B:1), and even rows show the residual maps obtained by averaging residuals over all bands.

across temporal nodes. In contrast, the proposed coupled strategy concatenates all temporal abundance tensors into a unified abundance domain and reconstructs them using a single untrained Transformer, preserving shared spatial structures while allowing temporal–spectral variability. As shown in Table V, the uncoupled strategy suffers from severe degradation, indicating that optimizing multiple independent Transformers is unstable under large cloud coverage. The proposed coupled strategy achieves better performance with fewer parameters and lower GPU memory, suggesting that the coupled representation balances cross-temporal structural consistency, time-specific flexibility, and optimization stability.

3) Contributions of Spatial Prior and Spectral Prior: To evaluate the contributions of the spatial prior introduced by the untrained Transformer and the spectral prior introduced by the spectral-guided deep decoder, three settings are considered on the *mountain* dataset: 1) without the spatial prior; 2) without the spectral prior; and 3) the complete C-Transformer. For the first two settings, the corresponding abundance tensor or spectral signatures are directly treated as learnable parameters and optimized by Adam. As shown in Table VI, the spatial prior achieves substantial improvements in reconstruction performance with acceptable computational complexity. The spectral prior further improves all reconstruction metrics, especially MSAM, with almost no additional computational cost. These results demonstrate the effectiveness of the two priors in improving reconstruction performance while maintaining reasonable computational complexity.

4) Influence of Reference Images: To evaluate the use of multitemporal reference information, we vary the number of

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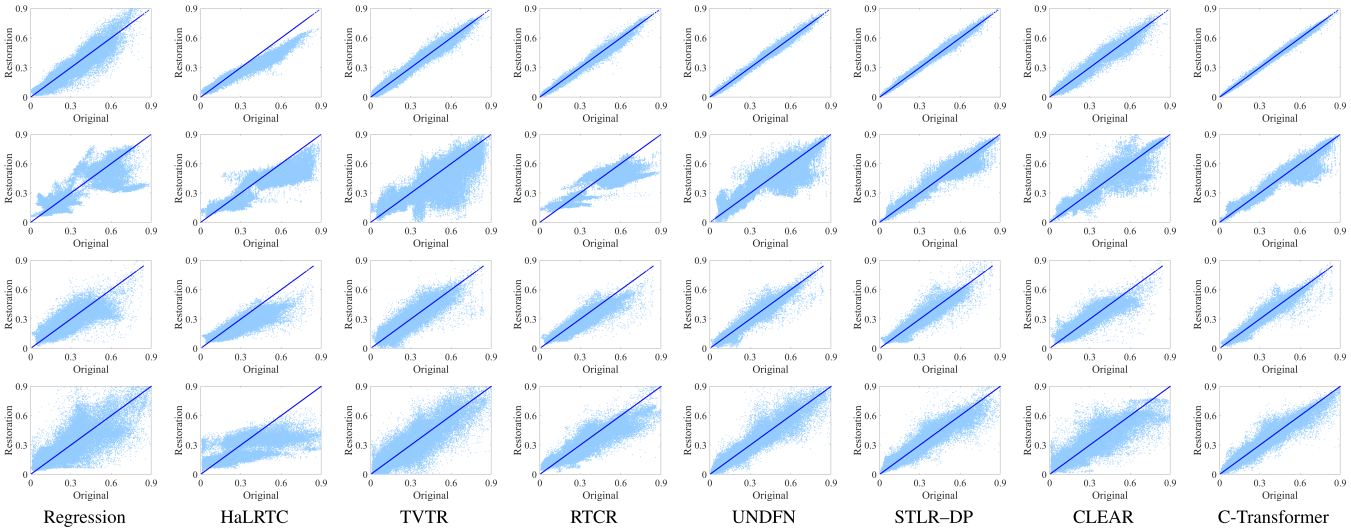


Fig. 8. Scatter plots comparing the original and reconstructed pixel values in the missing areas for different simulated datasets. The datasets are shown from top to bottom as follows: *mountain* at cloud-contaminated node 1, *farmland* at cloud-contaminated node 2, *city* at cloud-contaminated node 3, and *village* at cloud-contaminated node 4.

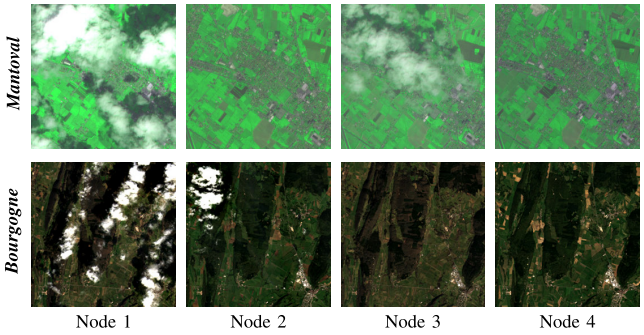


Fig. 9. Visual display of MTRSIs for the real datasets *Mantoval* (R:3, G:4, B:2) and *Bourgogne* (R:4, G:3, B:2).

TABLE VI

CONTRIBUTIONS OF SPATIAL PRIOR AND SPECTRAL PRIOR. THE BEST RESULTS ARE IN BOLD

Metric	w/o Spatial Prior	w/o Spectral Prior	Proposed
MPSNR	39.442	46.267	**47.455**
MSSIM	0.9686	0.9905	**0.9930**
MSAM	0.6823	0.4089	**0.3463**
Params	1.50M	**0.99M**	1.18M
GFLOPs	**0.006**	48.951	48.952

TABLE VII

QUANTITATIVE COMPARISON FOR THE EFFECT OF THE NUMBER OF REFERENCE IMAGES ON THE PROPOSED METHOD ON THE MOUNTAIN DATASET

Contaminated nodes	Metric	Observed	Number of reference images		
			0	1	2
Node 1	PSNR	7.322	44.009	44.5872	**45.225**
	SSIM	0.6992	0.9877	0.9888	**0.9903**
	SAM	11.7708	0.5026	0.4855	**0.4602**
Node 2	PSNR	7.501	46.200	46.6822	**47.406**
	SSIM	0.6810	0.9916	0.9923	**0.9935**
	SAM	10.6127	0.4110	0.4167	**0.3804**
Node 3	PSNR	9.578	48.123	49.0846	**49.734**
	SSIM	0.8014	0.9936	0.9944	**0.9954**
	SAM	6.7958	0.2190	0.2203	**0.1985**
Average	PSNR	8.134	46.111	46.7847	**47.455**
	SSIM	0.7272	0.9910	0.9918	**0.9930**
	SAM	9.7264	0.3775	0.3742	**0.3463**

cloud-free reference images from 0 to 2 on the *mountain* dataset, with the results reported in Table VII. As the number of reference images increases, PSNR and SSIM consistently improve while the average SAM decreases, indicating that additional temporal information further benefits structural and spectral recovery. Even without cloud-free references, C-Transformer substantially improves the contaminated observations, demonstrating its inherent reconstruction capability. These results demonstrate that C-Transformer can effectively exploit limited reference information and remains robust under reference-scarce scenarios.

5) Influence of Parameter: This section analyzes the influence of the number of latent abundance components r. As shown in Fig. 12, the PSNR values of the reconstruction results obtained by C-Transformer on the simulated datasets vary with different r values. Specifically, PSNR exhibits an upward trend as r increases within a specific range, subsequently leveling off. Based on this observation, it is recommended to select r values from the candidate set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

6) Convergence Analysis: We numerically demonstrate the convergence of the Adam-based solver developed in this work. Fig. 13 presents the training loss curves over iterations for various simulated datasets. As illustrated in Fig. 13, we observe that the training loss monotonically decreases and gradually

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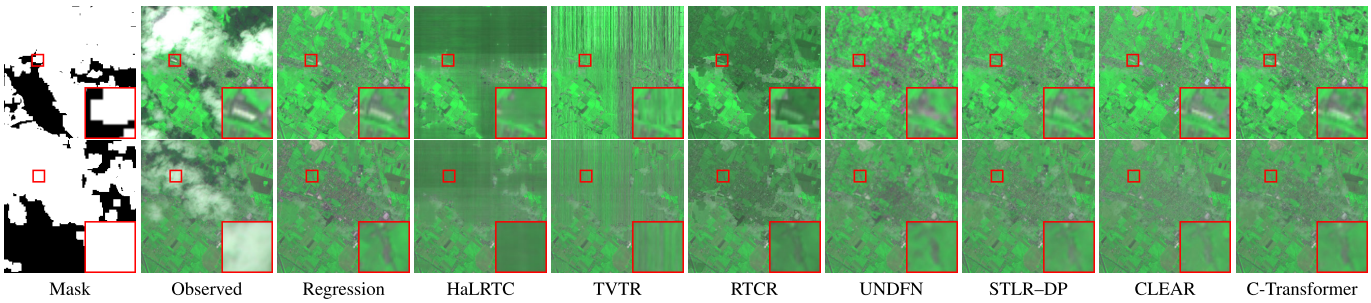


Fig. 10. Experimental results of various cloud removal methods on the *Mantoval* dataset. The images are presented from top to bottom at contaminated nodes 1 and 3.

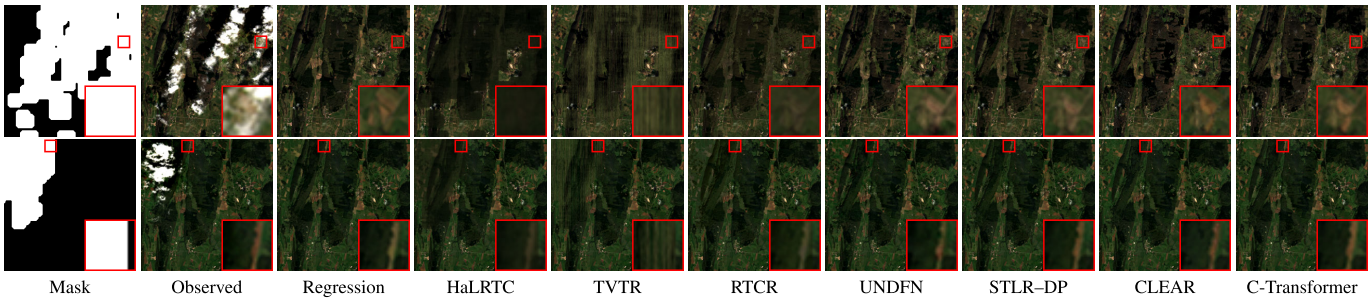


Fig. 11. Experimental results from various cloud removal methods on the *Bourgogne* dataset. The images are presented from top to bottom at contaminated nodes 1 and 2.

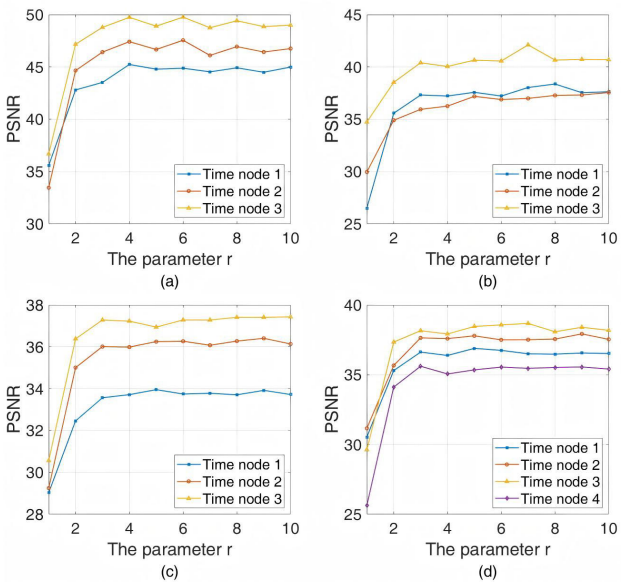


Fig. 12. PSNR values of the reconstruction results achieved by C-Transformer with varying r across various simulated datasets. (a) *Mountain*. (b) *Farmland*. (c) *City*. (d) *Village*.

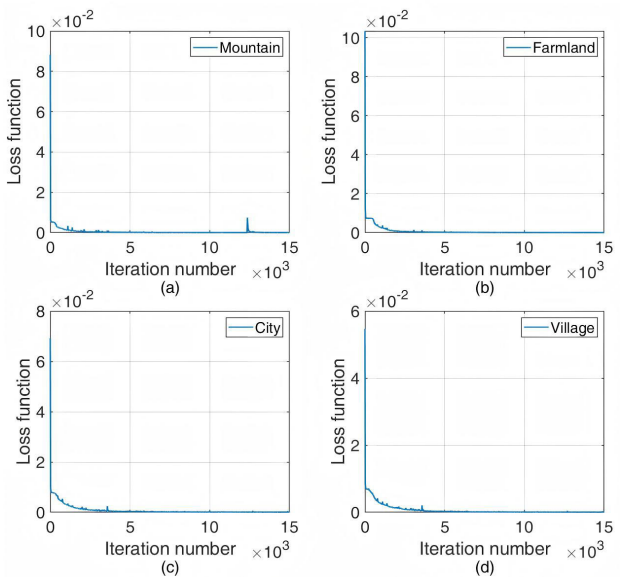


Fig. 13. History of loss function values across iteration counts for various simulated datasets. (a) *Mountain*. (b) *Farmland*. (c) *City*. (d) *Village*.

approaches zero as the number of iterations increases. This empirical evidence confirms the convergence of the proposed Adam-based solver.

VI. Conclusion

In this article, we proposed C-Transformer, an untrained coupled Transformer framework for MTRSI thick cloud removal. C-Transformer couples the abundance maps across temporal observations to exploit their structural information and employs an untrained Transformer to capture long-range spatiotemporal dependencies, while spectral-guided deep decoders independently represent time-specific spectral signatures by exploiting the available spectral information. Compared with existing unsupervised methods, C-Transformer provides stronger long-range representation capability, while compared with supervised methods, it offers greater flexibility across different MTRSIs without relying on external training data. Extensive experiments on simulated and real-world datasets demonstrate that C-Transformer consistently achieves superior reconstruction performance, with clearer

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spatial structures, better detail preservation, and improved spectral fidelity.

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